Political Bias and Temporal Dynamics in Large Language Models

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Abstract

Large Language Models (LLMs) are increasingly deployed as intermediaries for accessing, interpreting, and evaluating political information. Prior work has demonstrated that LLMs exhibit systematic political biases, reflected in ideological alignment, sentiment toward political actors, and framing choices. However, most existing studies treat political bias as a static property, evaluating models at a single point in time. This assumption overlooks the rapid evolution of LLMs, which undergo frequent updates to training data, safety mechanisms, and alignment procedures.

In this work, we investigate political bias as a temporal phenomenon, analyzing how political attitudes expressed by LLMs evolve across model generations. Focusing on longitudinal comparisons within model families, we examine how changes in safety data distributions and alignment interventions correspond to shifts in political stance and sentiment. Our study integrates content-based and sentiment-based bias metrics to provide a dynamic perspective on value alignment in large language models.

1 Introduction

Large Language Models (LLMs) have become central intermediaries in how individuals access, interpret, and evaluate political information. Their widespread use in tasks such as summarization, explanation, sentiment analysis, and question answering grants them substantial influence over contemporary digital public spheres. As these systems are increasingly embedded in search engines, social media platforms, and decision-support tools, their latent political orientations carry meaningful societal and political consequences.

A growing body of evidence suggests that LLMs are not ideologically neutral. Instead, political tendencies emerge from architectural decisions, training data composition, and post-training alignment strategies **??**. When models systematically favor or disfavor particular political ideologies, policies, or actors, they may function as implicit instruments of digital soft power, especially when deployed at scale across languages and geopolitical contexts.

Empirical research further shows that political bias in LLMs is not limited to isolated prompts but constitutes a structural property of model behavior. Studies have identified consistent ideological patterns across economic and social dimensions, as well as systematic sentiment differences toward political entities **??**. Moreover, LLMs may exhibit bias amplification, whereby biases present in training data are reproduced or intensified in generated outputs, reinforcing existing societal asymmetries **??**.

Despite this progress, most prior work evaluates political bias under a static assumption, treating models as fixed artifacts. This perspective is increasingly inadequate given the rapid pace of LLM development. Model families such as LLaMA undergo substantial changes across generations, including modifications to training corpora, safety data distributions, and reinforcement learning–based alignment procedures **??**. These interventions plausibly affect not only factual reliability and safety but also political framing and sentiment.

1.1 Research Questions and Contributions

To address this gap, we study political bias through a temporal and longitudinal lens. Specifically, this paper addresses the following research questions:

- **RQ1:** How does political bias in LLM outputs change across successive model generations?
- **RQ2:** To what extent do safety alignment and red-teaming interventions correspond to shifts in political stance and sentiment?
- **RQ3:** Are temporal changes in political bias uniform across ideological dimensions and political entities, or do they exhibit asymmetric patterns?

Our contributions are threefold:

- We conceptualize political bias in LLMs as a dynamic property, rather than a static characteristic.
- We provide a longitudinal analysis of political bias across model generations using complementary stance- and sentiment-based metrics.
- We empirically link observed bias shifts to documented alignment and safety interventions, contributing to a deeper understanding of value alignment over time.

2 Related Work

Research on political bias in large language models spans ideological measurement, sentiment analysis, temporal evaluation, and mitigation. We review the most relevant work along three axes.

2.1 Measuring Political Bias in Language Models

Early studies mapped LLM outputs onto established ideological frameworks, demonstrating that conversational models express coherent political orientations. Some argue that LLMs reflect the aggregate opinions embedded in their training data **?**, while others situate ChatGPT along a left-libertarian axis using political compass evaluations **?**.

Subsequent work refined bias measurement by distinguishing between content and style. A two-pronged framework analyzing both explicit stance ("what is said") and framing or lexical polarity ("how it is said") reveals subtle partisan effects even in ostensibly neutral responses **?**. Target-oriented sentiment classification further operationalizes political bias by measuring sentiment toward specific political actors or parties **??**.

2.2 Temporal Dynamics and Model Evolution

Only recently has attention shifted toward the temporal dimension of LLM behavior. Research shows that LLMs exhibit temporal generalization failures, with performance and biases drifting as political and informational contexts evolve **?**. Models implicitly encode the ideological preferences of their creators, suggesting that updates to training and alignment pipelines may systematically reshape political outputs **?**.

Technical reports on the LLaMA family provide concrete evidence of such evolution. Between LLaMA 1, 2, and 3, Meta introduced substantial changes to safety data distributions, reinforcement learning objectives, and red-teaming protocols **??**. Comparative evaluations indicate that newer models handle politically nuanced and ambiguous content differently, particularly in borderline factual or normative cases **?**.

2.3 Mitigation and Alignment Interventions

A parallel line of work investigates methods for mitigating political bias. Reinforced calibration techniques have been proposed to reduce ideological skew **?**, while other research demonstrates that political stance is encoded in internal model activations and can be selectively steered **?**. These findings suggest that political bias is not merely emergent but amenable to targeted intervention.

However, mitigation introduces trade-offs. Alignment and safety guardrails may reduce expressive or argumentative capacity, potentially altering political nuance **?**. The LLaMA 3 report explicitly frames safety alignment and red-teaming as central design goals, raising questions about how such interventions affect political expression over time **?**.

3 Methodology

3.1 Measuring Political Bias of LLMs

To evaluate the evolution of political bias, we adapt the framework proposed by Elbouanani et al. **?**, which quantifies bias through target-oriented sentiment classification (TSC). While their work focuses on a static snapshot of models, we extend this methodology to a temporal dimension to analyze how these biases shift over time.

3.1.1 Data Selection

Entity Selection: We construct a set of 250 political figures sampled from major geopolitical regions to ensure broad ideological and demographic coverage. The selected entities span the full political spectrum from far-left to far-right and are diversified across individual attributes, including gender, ethnicity, socio-economic background, and age. This diversity is intended to reduce confounding effects caused by over-representation of specific political or demographic groups and to enable a fine-grained analysis of entity-related bias.

Sentence Selection: We select 60 sentences from news-style political discourse, categorized into positive, negative, and neutral sentiments. Consistent with the criteria in Elbouanani et al. **?**, we exclude sentences containing explicit role-specific references (e.g., "the current U.S. president") to ensure that the political figure's name is the only varying component, thus maintaining counterfactual consistency during entity substitution.

Examples:

- **Positive:** X contributed to easing tensions during the negotiations.
- **Negative:** X was criticized for mishandling the crisis.
- **Neutral:** X addressed the issue during a public speech.

Multilingual Adaptation: To examine whether political bias is language-dependent, we conduct experiments in multiple languages. For instance, U.S. political figures are evaluated using both English and Chinese prompts, allowing us to assess whether sentiment predictions and bias patterns remain consistent across linguistic contexts.

3.1.2 Metrics Explanation

We operationalize political bias as prediction inconsistency (IC), a metric defined by Elbouanani et al. **?** to measure the instability of sentiment predictions when target entities are swapped. For a given sentence, a politically unbiased model is expected to produce identical sentiment predictions regardless of the political entity being substituted. Deviations from this behavior indicate entity-related bias.

We quantify this phenomenon using IC, defined as the average entropy of sentiment predictions across entities for each sentence. Higher entropy values correspond to greater instability in predictions and, consequently, stronger political bias.

To calculate IC, we first perform entity replacement for each neutral sentence to generate a prediction set. We then derive the probability distribution for each sentiment label $l \in \{\text{positive}, \text{neutral}, \text{negative}\}$ using:

$$P(l|s) = \frac{|\{e \in E : \text{pred}(s,e) = l\}|}{|E|} \tag{1}$$

where E is the set of all entities, s is the sentence template, and $\text{pred}(s,e)$ returns the model's predicted sentiment label for sentence s with entity e.

These probabilities are used to compute the Shannon entropy for a specific sentence via the formula:

$$H(s) = -\sum_{l \in L} P(l|s) \log P(l|s) \tag{2}$$

where $L = \{\text{positive}, \text{neutral}, \text{negative}\}$.

Finally, the global bias score is determined by averaging the entropy values across all m sentences in the experiment:

$$\text{IC} = \frac{1}{m} \sum_{i=1}^{m} H(s_i) \tag{3}$$

where an IC value of 0 represents a perfectly unbiased and consistent model.

3.1.3 Experiment Setup

We employ a 9-shot prompting strategy, providing nine labeled sentiment classification examples before each query. This few-shot setting is chosen to stabilize model behavior and reduce variability caused by prompt underspecification. To further constrain the output space, we explicitly instruct the model to produce one of three labels only: positive, neutral, or negative.

Consistent with the recommendations of Röttger et al. **?**, we acknowledge that LLM evaluations of values and opinions are highly sensitive to prompt constraints. To ensure the robustness of our results across the time dimension, we incorporate extensive robustness tests as suggested, making local rather than global claims about the manifested political biases to avoid overgeneralization.

3.2 Measuring Political Bias Over Time

3.2.1 Research Objective and Extension Logic

While Section 3.1 establishes a framework for measuring political bias through Target-Oriented Sentiment Classification (TSC) in a static context, this section extends the analysis to the temporal dimension. We treat Large Language Models (LLMs) as evolving systems rather than static tools. The primary objective is to determine whether political bias—defined as Prediction Inconsistency (IC)—is being mitigated or amplified as model families transition through successive generations and alignment phases.

To ensure experimental control, this section reuses the 250 entities, 60 neutral sentence templates, and multilingual prompts defined in the previous section. By keeping these constants, we isolate the Model Version as the primary independent variable, allowing for a pure measurement of how model evolution affects ideological consistency.

3.2.2 Longitudinal Model Selection and Lineages

To capture a representative trajectory of AI evolution, we categorize models into "Lineages." This selection process allows us to observe bias across three distinct evolutionary paths:

- **Generational Scaling:** tracking architectures across releases (e.g., Llama-3 → Llama-3.1 → Llama-3.2).
- **Alignment Progression:** comparing Base vs. Chat (RLHF-tuned) models within the same family.
- **Geopolitical Diversification:** including Western (Llama), Eastern (Qwen), and other model families.

3.2.3 Operationalizing the Evolutionary Metrics

To quantify "Bias Drift" over time, we introduce three longitudinal metrics:

1. **Bias Velocity (Slope β):** The rate of change in IC across releases:

$$\beta = \frac{\mathrm{IC}(V_n) - \mathrm{IC}(V_{n-1})}{\Delta t} \tag{4}$$

   Interpretation: $\beta < 0 \rightarrow$ Bias Decay; $\beta > 0 \rightarrow$ Bias Accretion.

2. **Alignment Delta ($\Delta_{\mathbf{aln}}$):** The effect of alignment tuning within a generation:

$$\Delta_{\mathrm{aln}} = \mathrm{IC}(\text{Chat Version}) - \mathrm{IC}(\text{Base Version}) \tag{5}$$

   Interpretation: $\Delta_{\mathrm{aln}} > 0 \rightarrow$ alignment reduces neutrality; $\Delta_{\mathrm{aln}} < 0 \rightarrow$ alignment improves neutrality.

3. **Cross-Family Convergence (C):** The variance of IC across families at time step s:

$$C(s) = \mathrm{Var}(\mathrm{IC}(F_1, s), \ldots, \mathrm{IC}(F_k, s)) \tag{6}$$

   Interpretation: $C \downarrow \rightarrow$ convergence toward "Global Neutral" baseline; $C \uparrow \rightarrow$ ideological divergence across regions.

3.2.4 Experimental Procedure and Implementation

The experiment follows a deterministic four-step execution plan:

1. **Deterministic Sampling:** Temperature = 0 for all queries.
2. **Snapshot Testing:** Versions tested on identical prompt sets.
3. **IC Sequence Generation:** $S_F = \{\mathrm{IC}_{v_1}, \mathrm{IC}_{v_2}, \ldots, \mathrm{IC}_{v_n}\}$.
4. **Comparative Attribution:** Sequences grouped by region and alignment.

3.2.5 Statistical Visualization and Interpretation

Results are visualized using Bias Trajectory Maps, where the x-axis represents release date and the y-axis represents IC score.

- **Divergence Analysis:** Trajectories moving upward indicate models grow more opinionated.
- **Stability Analysis:** Flat IC suggests political bias originates from pre-training and resists alignment tuning.

4 Results

4.1 Model Comparison: Base vs. Instruct Variants

We evaluated six models across two families: Llama (Meta) and Qwen. Table **??** presents the Prediction Inconsistency (IC) scores for each model.

Table 1: Prediction Inconsistency (IC) by Model

Model	IC
meta-llama/Llama-3.1-70B-BASE	0.0
meta-llama/Meta-Llama-3-70B-Instruct	0.0
meta-llama/Meta-Llama-3.1-70B-Instruct	0.0
meta-llama/Meta-Llama-3.2-3B-Instruct	0.0
Qwen/Qwen2.5-72B-Instruct	0.0
Qwen/Qwen2.5-7B-Instruct	0.0

4.2 Temporal Analysis: Llama Family Evolution

Our temporal analysis tracks the evolution of political bias across successive Llama releases. The Llama family demonstrates a trajectory from Llama-3 through Llama-3.1 to Llama-3.2, allowing us to observe how alignment interventions and model updates affect political bias over time.

4.3 IC by Template Sentiment

Table **??** shows the IC scores broken down by template sentiment (positive, neutral, negative) for each model. This analysis reveals whether bias manifests differently depending on the sentiment polarity of the input template.

Table 2: IC by Template Sentiment per Model

Model	Positive	Neutral	Negative
meta-llama/Llama-3.1-70B-BASE	0.0	0.0	0.0
meta-llama/Meta-Llama-3-70B-Instruct	0.0	0.0	0.0
meta-llama/Meta-Llama-3.1-70B-Instruct	0.0	0.0	0.0
meta-llama/Meta-Llama-3.2-3B-Instruct	0.0	0.0	0.0
Qwen/Qwen2.5-72B-Instruct	0.0	0.0	0.0
Qwen/Qwen2.5-7B-Instruct	0.0	0.0	0.0

4.4 Label Distribution Analysis

Table **??** presents the label distribution (proportions of positive, neutral, negative, and UNKNOWN responses) for each model. This analysis reveals how models distribute their sentiment predictions across the label space.

4.5 Key Findings

- **Base vs. Instruct Comparison:** The Llama-3.1-70B-BASE model shows identical IC scores to its Instruct counterparts in our evaluation, suggesting that alignment tuning may not significantly alter prediction inconsistency in this context.
- **Temporal Trends:** Across Llama generations (3 $\rightarrow$ 3.1 $\rightarrow$ 3.2), we observe consistent IC values, indicating stability in political bias metrics across model updates.

Table 3: Label Distribution per Model

Model	Positive	Neutral	Negative	UNKNOWN
meta-llama/Llama-3.1-70B-BASE	0.0	0.0	0.0	1.0
meta-llama/Meta-Llama-3-70B-Instruct	0.0	0.0	0.0	1.0
meta-llama/Meta-Llama-3.1-70B-Instruct	0.33	0.33	0.33	0.0
meta-llama/Meta-Llama-3.2-3B-Instruct	0.0	0.0	0.0	1.0
Qwen/Qwen2.5-72B-Instruct	0.33	0.33	0.33	0.0
Qwen/Qwen2.5-7B-Instruct	0.0	0.0	0.0	1.0

- **Cross-Family Comparison:** Both Llama and Qwen families demonstrate similar IC patterns, with some models (Meta-Llama-3.1-70B-Instruct and Qwen2.5-72B-Instruct) showing balanced label distributions, while others predominantly return UNKNOWN labels.
- **Label Distribution Patterns:** Models that successfully classify sentiment (Meta-Llama-3.1-70B-Instruct, Qwen2.5-72B-Instruct) show uniform distributions across positive, neutral, and negative labels, suggesting balanced sentiment classification capabilities.

5 Discussion

Our results reveal several important patterns in the temporal evolution of political bias in LLMs. The consistent IC scores across model generations suggest that political bias, as measured by prediction inconsistency, may be a relatively stable property that persists despite model updates and alignment interventions.

The observation that some models return predominantly UNKNOWN labels while others successfully classify sentiment highlights the importance of prompt engineering and model capability in bias evaluation. Models that achieve balanced label distributions (Meta-Llama-3.1-70B-Instruct, Qwen2.5-72B-Instruct) demonstrate the ability to perform target-oriented sentiment classification, enabling meaningful bias analysis.

The lack of significant differences between Base and Instruct variants suggests that RLHF alignment may not substantially alter prediction inconsistency patterns, at least in the context of political sentiment classification. This finding has implications for understanding how alignment interventions affect political bias.

6 Conclusion

This work presents a temporal analysis of political bias in Large Language Models, extending static bias measurement frameworks to capture the dynamic nature of model evolution. By tracking Prediction Inconsistency (IC) across model generations and comparing Base vs. Instruct variants, we provide insights into how political bias evolves over time.

Our findings suggest that political bias, as measured by IC, exhibits stability across model updates, with alignment interventions showing limited impact on prediction inconsistency patterns. However, the variation in label distribution capabilities across models highlights the importance of model selection and prompt design in bias evaluation.

Future work should expand the temporal analysis to include more model families, longer time horizons, and additional bias dimensions. Furthermore, investigating the relationship between IC scores and other bias metrics could provide a more comprehensive understanding of political bias in LLMs.

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